CLA 4 - ML Model Deployment Strategies

Dataset: Titanic - Machine Learning from Disaster

1. Online Inference vs Batch Inference

Online inference processes one record at a time. Batch inference processes multiple records together.

2. Streaming Inference vs Batch Inference

Streaming inference processes incoming records continuously, one at a time. Batch inference processes collected records together.

3. Why FastAPI is used for Model Serving

FastAPI provides REST API endpoints, automatic documentation and Pydantic validation for machine learning model serving.

4. Why Reuse the Same Trained Model

Reusing the same model ensures consistent predictions, fair comparisons and avoids unnecessary retraining.